

Supplementary Information for "Engineered Biopolymer Composite, an affordable carbon-negative alternative to concrete"

Barney H. Miao^{1*}, Andrew C. Lesh¹, Paul Hart³, Steven Song¹,
David J. Loftus², Michael D. Lepech¹

¹*Department of Civil and Environmental Engineering, Stanford University, 473 Via Ortega, Stanford, 94305, California, USA.

²Department , NASA Ames Research Center, De France Ave, Moffett Field, 94035, California, USA.

³nano@stanford, Stanford University, 348 Via Pueblo, Stanford, 94305, California, USA.

*Corresponding author(s). E-mail(s): barneym@stanford.edu;

Overview of document

This document presents the supplementary materials accompanying the article entitled "Engineered Biopolymer Composite, an affordable carbon-negative alternative to concrete." These materials provide additional details on the analytical methods and investigations conducted throughout the study. The document includes expanded discussions on the design considerations of the Engineered Biopolymer Composite (EBC), experimental testing procedures, microstructural characterization, as well as life cycle and economic analysis.

Design considerations for Engineered Biopolymer Composite

Engineered biopolymer composite (EBC) consists of densely packed aggregates held together by biopolymers that fill inter-particle voids, forming cohesive interparticle bridges between the aggregate particles. Similar to other biopolymer-bound composites and soils stabilized with bio-based binders.

The compressive strength of EBC is governed by the interfacial bond between the biopolymer binder and the aggregate, the mechanical properties of the bridges themselves, the strength of the aggregate, and the geometric arrangement of individual aggregate particles. Strategies to enhance the strength of these biocomposites include increasing the strength of the biopolymer binder through careful design (e.g., proportioning different polymers for optimal performance), as well as optimizing the level of compaction and the solvent used during fabrication [1].

In this study, the biopolymer binder used consists of a blend of lignin, obtained through the LignoForce process, and methyl cellulose. Various blends were investigated, with lignin serving as the primary binder and methyl cellulose functioning as an additive. To effectively utilize the biopolymer binder for manufacturing EBC, a solvent mixture consisting of 80% acetic acid and 20% deionized water was used to fully dissolve the biopolymer blend.

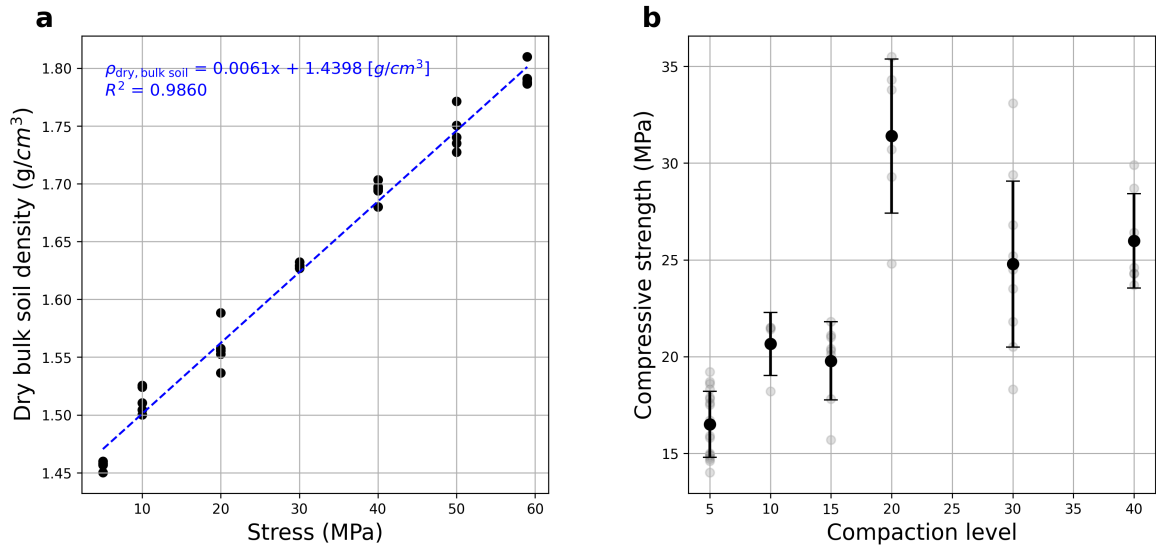
The solvent ratio was selected based on preliminary experiments, which showed that lignin-based composites exhibited the highest compressive strength at this composition (see Supplementary Figure 1). At lower acetic acid concentrations, incomplete dissolution of lignin was observed, with visible undissolved polymer. At concentrations above 80 %, no further improvement in compressive strength was noted. This was attributed to complete dissolution of lignin at acetic acid concentrations exceeding 80 %, while lower concentrations failed to fully solubilize the polymer. Undissolved lignin does not actively contribute to binding the aggregates within the composite, thus reducing overall strength. Based on these findings, a solvent mixture of 80 % acetic acid and 20 % deionized water was used throughout this study [2].

Although pure acetic acid effectively dissolves lignin, mixtures of acetic acid and water have been shown to be sufficient. Prior studies have demonstrated that lignin can be effectively dissolved using aqueous mixtures with organic solvents (e.g., dioxane) [3]. Moreover, the use of mixed solvents reduces both the financial cost and carbon footprint associated with lignin-based biocomposite production, enhancing their feasibility for practical applications [2, 4].

The strength of biopolymer-bound soil composites depends heavily on two design parameters: dry bulk soil density and biopolymer content [1]. Dry bulk soil density refers to the density of the granular material in the finished composite, while biopolymer content refers to the amount of biopolymer binder present in the biocomposite. Higher levels of both dry bulk soil density and biopolymer content are associated with increased physical strength in the composite material. Therefore, previous design

considerations for lignin biocomposites have primarily focused on these two parameters. Similarly, in the design of engineered biopolymer composites, we considered these same two parameters in developing a design guide.

Dry bulk soil density is influenced by the level of compaction applied during material preparation. To increase the dry bulk density of Grade 90 silica sand used in the composite, higher compaction levels can be employed (see Supplementary Fig. 1a). A relationship relating double-sided compaction level and dry bulk soil density was found to be $\rho_{\text{dry, bulk soil}} = 0.0061 C_{\text{level}} + 1.44 \text{ [g/cm}^3\text{]}$. However, increasing the bulk density also reduces the total void space within the material, which in turn limits the maximum volume of biopolymer that can be incorporated into the composite.



Supplementary Fig. 1: Dry bulk soil density of aggregates and compressive strength of EBC at varying levels of compaction. **a**, The dry bulk soil density of the aggregates within EBC increases linearly with the level of compaction. A linear fit was generated, allowing identification of the dry bulk soil density of this granular material at different compaction levels. **b**, The corresponding compressive strength of EBC increases slightly as the level of compaction increases. The optimal compaction level for EBC was found to be 20 MPa for this mix design. At higher levels of compaction, the compressive strength no longer increases significantly; therefore, 20 MPa was selected as the optimal value.

To determine the optimal compaction level—defined as the point where most of the benefits of increased compaction are gained without significantly compromising the biopolymer content—we conducted an optimization study. The results revealed a characteristic hump-shaped relationship: the compressive strength of the engineered biopolymer composite (EBC) increases with compaction, peaking at approximately 20 MPa under double-sided compaction (see Supplementary Fig. 1b). Beyond this point, further compaction leads to a reduction in compressive strength. Although the soil density

continues to increase, the reduced biopolymer content negatively affects the overall mechanical performance of the composite. Thus, for this study, we manufactured EBC using double-sided compaction at 20 MPa, corresponding to a dry bulk soil density of $\rho_{\text{dry, bulk soil}} = 1.56 \text{ g/cm}^3$.

Manufactured Engineered Biopolymer Composite Mixes

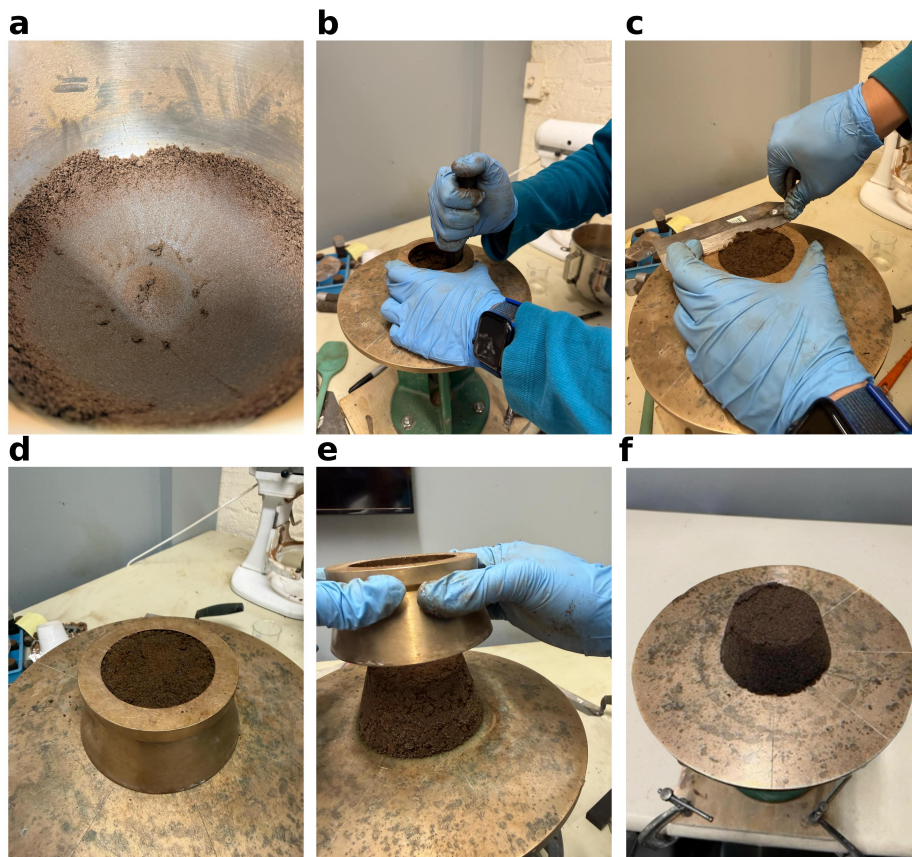
To investigate the influence of lignin content and methyl cellulose content on the material properties of the Engineered Biopolymer Composite, 25 mixes with varying biopolymer contents were manufactured. For each mix, at least 8 specimens were produced. The manufactured materials are detailed in Supplementary Table S1.

Supplementary Table S1: Mix designs of Engineered Biopolymer Composite investigated in study

Mix number	% Lignin	% Methyl cellulose	Compressive strength (MPa)	Density (g/cm ³)	Modulus (MPa)	Dry bulk soil density (g/cm ³)
1	8.3	0.0	5.24 ± 0.72	1.65 ± 0.01	389 ± 51	1.53 ± 0.01
2	16.1	0.0	8.20 ± 0.66	1.87 ± 0.01	508 ± 110	1.61 ± 0.01
3	7.5	0.8	10.08 ± 0.66	1.69 ± 0.01	889 ± 84	1.56 ± 0.01
4	14.4	1.6	13.57 ± 1.30	1.85 ± 0.02	1283 ± 365	1.59 ± 0.02
5	6.7	1.7	15.23 ± 0.58	1.69 ± 0.01	1217 ± 165	1.56 ± 0.01
6	12.8	3.2	19.80 ± 1.34	1.84 ± 0.01	2142 ± 328	1.58 ± 0.01
7	5.8	2.5	20.03 ± 1.48	1.71 ± 0.01	1444 ± 198	1.58 ± 0.01
8	11.2	4.8	24.01 ± 3.16	1.88 ± 0.02	2194 ± 485	1.59 ± 0.01
9	11.0	2.8	23.47 ± 0.75	1.80 ± 0.06	1850 ± 394	1.58 ± 0.05
10	12.1	3.0	31.40 ± 3.60	1.83 ± 0.06	1891 ± 278	1.59 ± 0.06
11	9.4	1.7	13.72 ± 1.59	1.78 ± 0.00	1059 ± 80	1.60 ± 0.00
12	9.9	3.3	23.45 ± 2.81	1.79 ± 0.01	1632 ± 203	1.58 ± 0.01
13	12.8	3.2	16.82 ± 2.62	1.81 ± 0.02	1568 ± 252	1.56 ± 0.01
14	15.2	3.1	17.73 ± 3.95	1.83 ± 0.01	1244 ± 318	1.55 ± 0.01
15	10.0	4.3	26.21 ± 2.54	1.84 ± 0.02	2136 ± 376	1.61 ± 0.01
16	10.0	6.4	35.39 ± 0.51	1.81 ± 0.01	2542 ± 496	1.55 ± 0.01
17	6.7	0.0	4.48 ± 1.36	1.67 ± 0.03	261 ± 94	1.56 ± 0.03
18	6.7	0.5	5.96 ± 0.80	1.70 ± 0.02	397 ± 103	1.58 ± 0.02
19	6.7	1.0	10.25 ± 0.61	1.73 ± 0.03	984 ± 176	1.61 ± 0.03
20	6.7	3.0	23.24 ± 0.31	1.79 ± 0.03	1320 ± 308	1.62 ± 0.02
21	4.9	6.7	32.06 ± 1.77	1.77 ± 0.02	2879 ± 484	1.59 ± 0.02
22	6.7	6.7	24.77 ± 0.99	1.78 ± 0.01	1845 ± 348	1.57 ± 0.01
23	6.7	3.8	28.86 ± 1.78	1.74 ± 0.01	2561 ± 514	1.58 ± 0.01
24	8.4	3.6	26.11 ± 1.12	1.77 ± 0.02	2556 ± 380	1.58 ± 0.02
25	6.0	6.5	29.71 ± 2.53	1.81 ± 0.02	2520 ± 448	1.58 ± 0.02

Slump testing of Engineered Biopolymer Composite

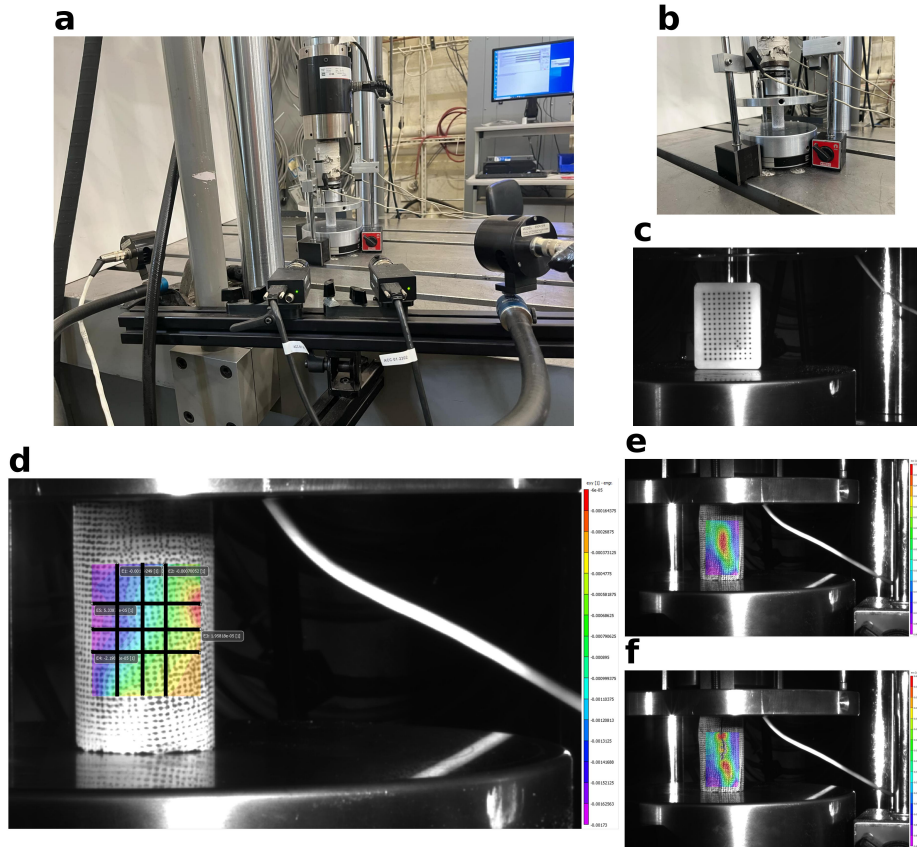
The flow table test (Supplementary Fig. 2) was used to assess the workability and consistency of the Engineered Biopolymer Composite under standardized conditions adapted from ASTM C1437. Freshly prepared mixtures were loaded into the ASTM flow mold in two equal layers, with each layer compacted by 20 tamps using the standard tamper to ensure uniform consolidation. After the mold was filled and the surface leveled flush with the top edge, the mold was carefully removed vertically to allow unrestricted spreading of the material. The flow table was then subjected to 25 drops within 15 s in accordance with ASTM C1437. The resulting spread diameter was measured along four orthogonal directions, and the average value was used to quantify the flowability of the composite mixture.



Supplementary Fig. 2: Flow table test of Engineered Biopolymer Composite following ASTM standards. **a**, Consistency of the Engineered Biopolymer Composite was evaluated using the flow table method following ASTM C1437 (Standard Test Method for Flow of Hydraulic Cement Mortar), using apparatus conforming to ASTM C230/C230M. This test measures the workability and flow behavior of mortar or similar cementitious mixtures and was adapted here for the composite material. **b**, The freshly mixed material is placed into the flow mold in two layers. Each layer is tamped 20 times with the tamper as specified in ASTM C1437 to consolidate the specimen. **c**, Excess material is struck off using a trowel so that the surface is flush with the top of the mold. **d**, The mold is carefully lifted vertically to allow the specimen to spread. **e**, The flow table is dropped 25 times within 15 s as specified by ASTM C1437. **f**, The flow is determined by measuring the average diameter of the spread specimen in four directions to quantify the consistency of the mixture.

Digital image correlation of EBC

To further characterize the elastic deformation behavior of the Engineered Biopolymer Composite (EBC), full-field, non-contact strain measurements were performed using stereo digital image correlation (DIC) during quasi-static uniaxial compression testing (Supplementary Fig. 3a). Cylindrical specimens with a diameter of 1 inch were prepared with a stochastic black-and-white speckle pattern to enable accurate surface tracking throughout deformation (Supplementary Fig. 3b). Compression tests were conducted using an MTS Criterion Model 43 at a displacement-controlled loading rate of 0.02 mm s^{-1} . Simultaneously, stereo image pairs were captured and synchronized with the applied load through timestamp matching, enabling direct correlation between mechanical response and full-field strain evolution.



Supplementary Fig. 3: Digital image correlation tests for characterizing the Young's modulus and Poisson's ratio of Engineered Biopolymer Composite. **a**, Experimental setup for stereo digital image correlation (DIC) during quasi-static uniaxial compression of Engineered Biopolymer Composite. **b**, Cylindrical specimen (1-inch diameter) prepared with a speckle pattern and subjected to compression; LVDTs and DIC measurements are used to determine strain and compute Young's modulus. **c**, Calibration target used for DIC system calibration. **d**, Representative virtual extensometer placement within the area of interest for measuring longitudinal and transverse strains used to calculate Poisson's ratio. **e**, **f**, Full-field strain distributions before and after failure, respectively, illustrating deformation localization in the specimen.

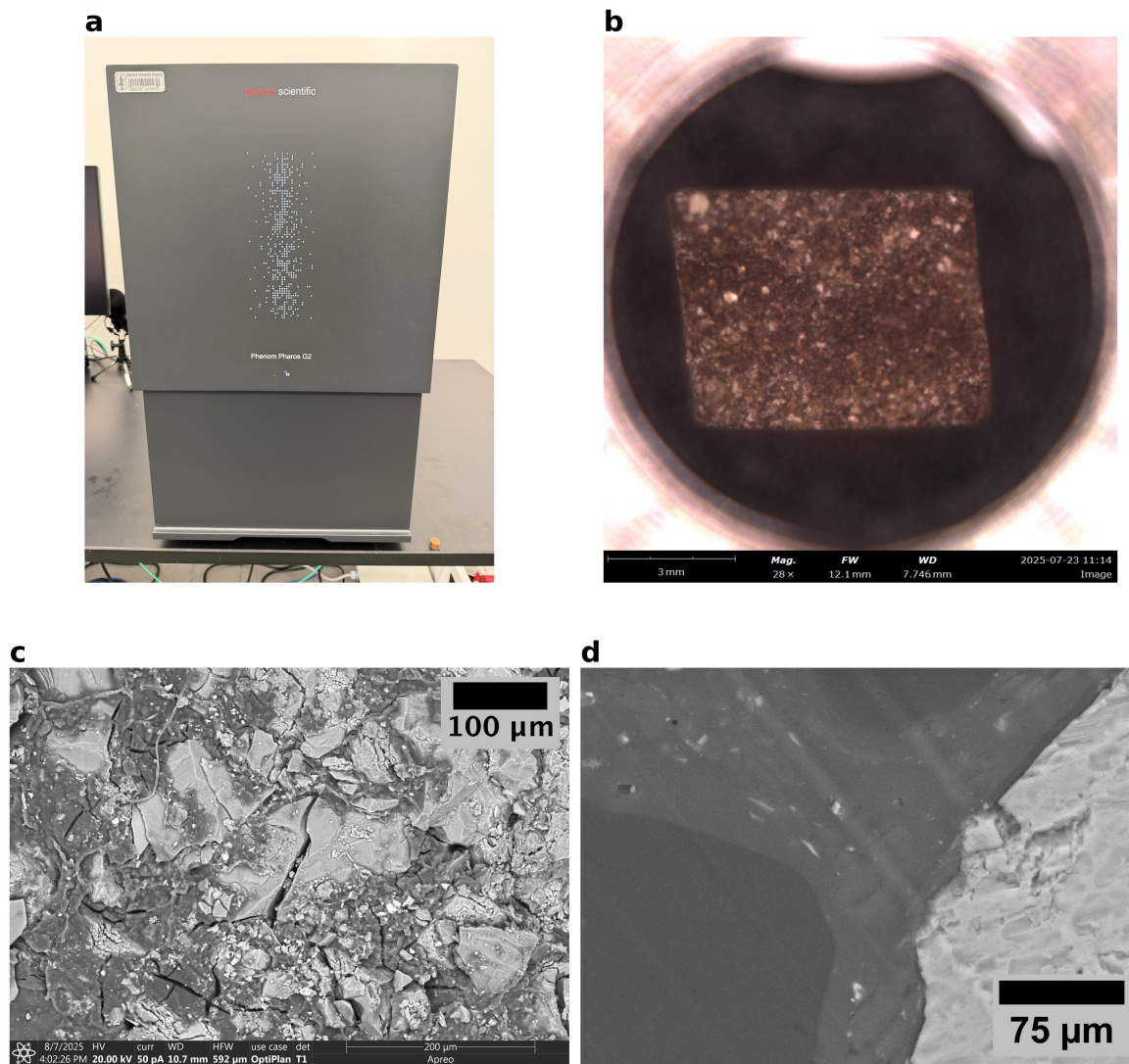
96 Prior to testing, the Vic-3D stereo DIC system was calibrated using a 3 mm dot-pitch calibration
 97 target (Supplementary Fig. 3c). Calibration images were acquired from at least ten orientations to
 98 ensure robust three-dimensional reconstruction accuracy, and calibration was accepted only when the
 99 reprojection error was below 0.05 pixels. Following acquisition, image sequences were processed in
 100 Vic-3D using the validated calibration file. A rectangular area of interest (AOI) encompassing the
 101 specimen gauge section was selected for correlation analysis using the default subset size. Within
 102 the AOI, virtual extensometers were defined to quantify both longitudinal and transverse strains
 103 (Supplementary Fig. 3d). Three longitudinal extensometers were positioned parallel to the loading
 104 direction at approximately 1/6, 1/2, and 5/6 of the specimen width to capture axial strain distribution,
 105 while three transverse extensometers located near the specimen midpoint measured lateral strain.
 106 Engineering strain obtained from DIC was synchronized with engineering stress data recorded by
 107 the MTS load cell. Young's modulus was determined from the slope of the linear elastic region of
 108 the engineering stress-strain response using linear regression, while Poisson's ratio was calculated as
 109 the negative ratio of transverse to longitudinal strain within the same elastic regime. In addition to
 110 quantitative elastic property measurements, DIC enabled visualization of full-field strain evolution and
 111 deformation localization throughout loading and failure (Supplementary Fig. 3e and Supplementary
 112 Fig. 3f), providing insight into crack initiation, strain concentration, and heterogeneous deformation
 113 behavior in the EBC specimens.

114 **Training approach for predicting compressive strength of** 115 **Engineered Biopolymer Composite**

116 A supervised deep learning model was developed to predict compressive strength as a function of
 117 lignin and methyl cellulose contents. A feedforward neural network with three hidden layers (64, 32,
 118 and 16 neurons) and ReLU activations was trained using the Adam optimizer with a learning rate of
 119 7.5×10^{-4} . The input features consisted of the mass fractions of lignin and methyl cellulose, while the
 120 target values were the measured ultimate compressive strengths of EBC. Both inputs and outputs
 121 were standardized using z-score normalization. The dataset was split into 80 % training and 20 %
 122 validation sets. Training was conducted over 150 epochs with a batch size of 16, and fixed random
 123 seeds were used to ensure reproducibility.

Scanning electron microscopy of EBC

Scanning electron microscopy was conducted to help visualize the microstructure of Engineered Biopolymer Composite, specifically how the biopolymer binder interacts with the aggregate particles in the material. Supplementary Fig. 4, shows the overview of how a Thermo Fischer Scientific Phenom Pharos desktop was used to visualize EBC specimens on the microscale.



Supplementary Fig. 4: Overview of scanning electron microscopy experiments conducted on Engineered Biopolymer Composite. **a**, Thermo Fisher Scientific Phenom Pharos desktop field-emission gun scanning electron microscope (FEG-SEM) used for imaging. **b**, Representative EBC specimen prepared for SEM analysis; samples were embedded in low-viscosity epoxy, cured, trimmed to a ~ 1 mm pyramidal block using a Leica TXP, and polished via ultramicrotomy with a 45° diamond knife. **c**, SEM image of an unpolished EBC specimen showing aggregate particles bound by the biopolymer matrix, with microcracks visible within both phases. **d**, High-magnification SEM image of a polished sample highlighting the interface between the biopolymer binder and aggregate, where localized delamination is observed. Imaging was performed under low-vacuum, low-kV conditions to enhance contrast between epoxy and biopolymer without conductive coating.

Statistical Descriptors of Composite Microstructure

To quantitatively characterize the spatial structure of the composite materials, we employed two statistical descriptors derived from stochastic geometry: the two-point probability function and the lineal path function. These metrics provide insight into phase distribution, connectivity, and interfacial relationships in multiphase materials. All descriptors were computed for each of the three constituent phases—aggregate, biopolymer, and void—using segmented 2D microstructural images.

The lineal path function, $L_i(r)$, represents the probability that a randomly placed line segment of length r lies entirely within phase i . It is particularly sensitive to phase continuity and connectedness along linear paths, making it useful for assessing potential transport or load-bearing pathways.

$$L_i(r) = \mathbb{P}(\text{a line segment of length } r \text{ lies entirely within phase } i)$$

Lineal path functions were computed for each phase:

- $L_{\text{agg}}(r)$: aggregate continuity,
- $L_{\text{bio}}(r)$: biopolymer continuity,
- $L_{\text{void}}(r)$: void continuity.

Computation was performed by scanning along discrete directions (e.g., horizontal and vertical) across the image, extending line segments and checking for uninterrupted sequences of the target phase. Results were averaged over many line placements to ensure statistical robustness.

Higher values of $L_i(r)$ indicate longer uninterrupted paths in phase i , reflecting its degree of spatial continuity. Comparison across materials provides insight into phase connectivity and heterogeneity.

The two-point probability function, $S_2^{(i,j)}(\mathbf{r})$, defines the probability that two points, separated by a vector $\mathbf{r}$, simultaneously belong to phases i and j , respectively:

$$S_2^{(i,j)}(\mathbf{r}) = \mathbb{P}(\mathbf{x} \in \text{phase } i, \mathbf{x} + \mathbf{r} \in \text{phase } j)$$

Assuming statistical homogeneity and isotropy, S_2 is a function of the scalar separation distance $r = \|\mathbf{r}\|$:

$$S_2^{(i,j)}(r) = \mathbb{P}(\mathbf{x} \in \text{phase } i, \mathbf{x} + \mathbf{r} \in \text{phase } j)$$

We computed the following two-point correlation functions for both materials:

- $S_2^{(\text{agg},\text{bio})}(r)$: aggregate–biopolymer correlation,
- $S_2^{(\text{agg},\text{agg})}(r)$: aggregate autocorrelation,
- $S_2^{(\text{void},\text{void})}(r)$: void autocorrelation.

These functions were estimated using pair-counting over labeled image data. Elevated values of $S_2^{(i,j)}(r)$ at short distances indicate a higher likelihood of spatial association between the corresponding phases, revealing information about intermixing, clustering, and phase separation.

Life cycle assessment

A cradle-to-gate life cycle assessment (LCA) was carried out using SimaPro software and the Impact 2002+ method. One of the primary reasons for selecting the Impact 2002+ method over others is its established use in performing LCAs on other soil-based building materials [4]. The Impact 2002+ method reports environmental impacts across 14 different characterization categories [5], covering human health, ecosystem quality, climate change, and resource depletion.

Equation 1 was used to calculate the environmental impacts of manufacturing lignin biocomposite. The total impact of lignin-based BSC can be attributed to four main activities: transportation requirements, electricity use, solvent use, and lignin production. For this LCA, we do not consider the additional footprint incurred by transporting the manufactured lignin biocomposite to the application site (i.e., a cradle-to-gate approach was used). The LCA was conducted using the Impact 2002+ method in SimaPro. Equation 2 shows the general form used to calculate the environmental impacts of lignin-based BSC, where f is the impact per unit, m is in kilograms (kg), E is in kilowatt-hours (kWh), and T is in tonne-kilometres (tkm).

$$Total\ Impact = Transportation_{Imp.} + Electricity_{Imp.} + Solvent_{Imp.} + Biopolymer_{Impact} \quad (1)$$

where $Transportation_{Imp.}$ represents the environmental impact associated with transporting raw materials to the manufacturing site, $Electricity_{Imp.}$ is the impact from electricity consumption during processing, $Solvent_{Imp.}$ denotes the impact associated with solvent production and use, and $Biopolymer_{Impact}$ represents the environmental impact attributed to lignin production and sourcing.

Transporting biopolymer binder to sites of manufacture

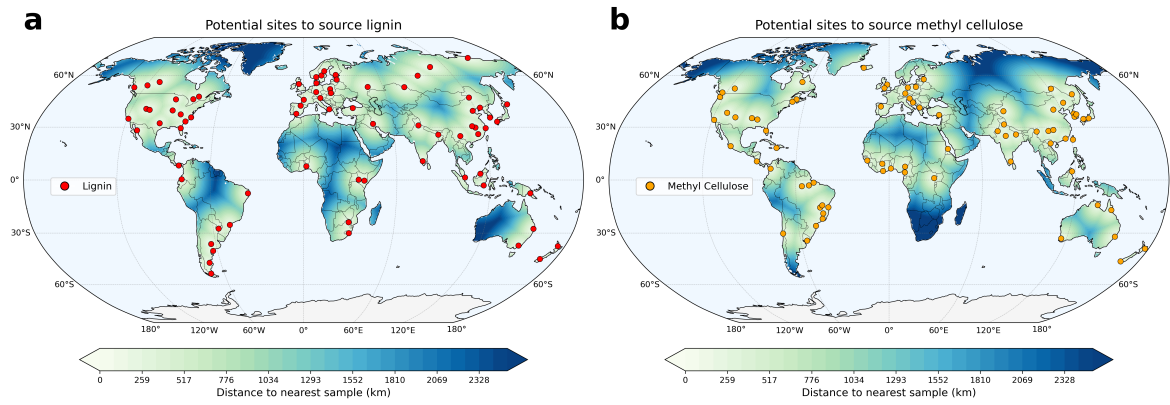
The impact of transportation, $Transportation_{Imp.}$ (Equation 2), accounts for the transportation of the constituents (solvent, lignin, and aggregate) of Engineered Biopolymer Composite to the manufacturing site. To calculate this impact, distances between the manufacturing site and the locations where lignin and methyl cellulose can be potentially sourced (see Supplementary Figure 5) were obtained [6]. The transportation distance for the solvents was estimated based on an average distance between chemical plants and manufacturing sites and was multiplied by the mass of solvent used

(acetic acid). For this life cycle assessment, it is assumed that the water used in the mixed solvent is sourced directly from the manufacturing site.

$$Transportation_{Imp.} = f_{transportation} * (m_{acid} * T_{acid} + m_{Lig} * T_{Lig} + m_{MC} * T_{MC}) \quad (2)$$

where $f_{transportation}$ is the transportation emission factor (e.g., kg CO₂e per ton-km), m_{acid} , m_{Lig} , and m_{MC} represent the masses of acetic acid, lignin, and methyl cellulose, respectively, and T_{acid} , T_{Lig} , and T_{MC} denote the corresponding transportation distances from the source locations to the manufacturing site.

The distances calculated from Supplementary Figure 5 were used to compute the global carbon footprint distribution of EBC. However, for the results presented in Figure 4c–n of the main manuscript, a representative transportation distance of approximately 1000 km was used, as this was the most frequently occurring transport distance from sources of lignin and methyl cellulose in the dataset.

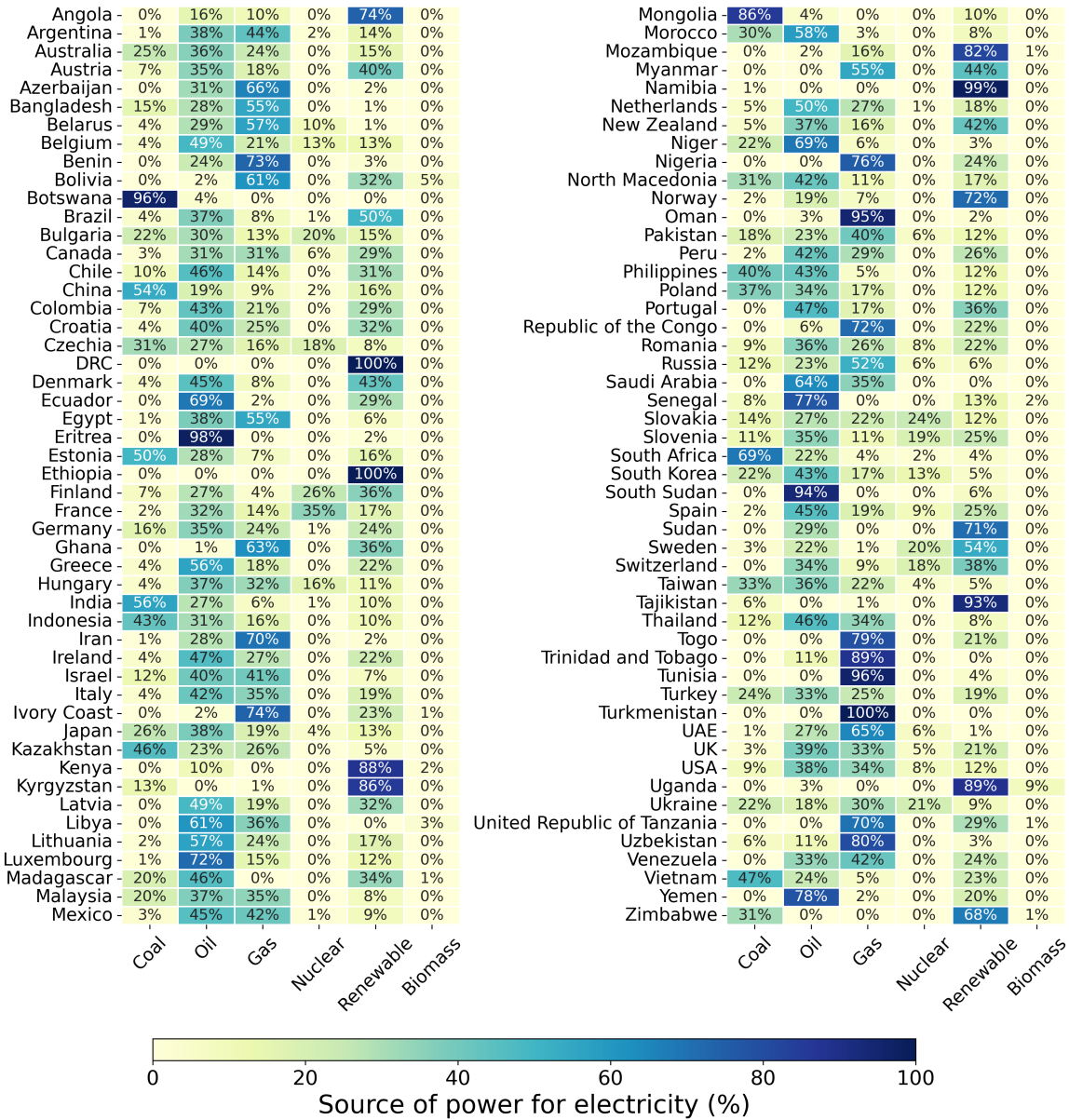


Supplementary Fig. 5: Potential sites and distances from potential sites where lignin and methyl cellulose can be sourced to source lignin and methyl cellulose biopolymers. a and b: Distances between manufacturing locations and potential sources of lignin and methyl cellulose were determined to estimate the carbon footprint associated with the production of the Engineered Biopolymer Composite.

Energy use during manufacture of Engineered Biopolymer Composite

The impact of energy use, $Electricity_{Imp.}$ (Equation 3), accounts for the electricity consumption associated with the fabrication of Engineered Biopolymer Composite — that is, the physical mixing of constituents and the energy required to desiccate the wet mix. During fabrication, lignin and methyl cellulose were first dry blended in an electrically powered stand mixer for approximately 2.5 minutes. Granular material was then added, and the mixture was further dry mixed for an additional 2.5 minutes to achieve an uniform dry blend. The solvent is added to the dried mix and mixed for 5 minutes to obtain the wet mix. During desiccation an electrically powered oven (at 40 C°) is used to accelerate the desiccation of the composite for 5 days.

$$Electricity_{Imp.} = f_{electricity} * \left(\frac{E_{fabrication} + E_{desiccation}}{\eta} \right) \quad (3)$$



Supplementary Fig. 6: Breakdown of sources for electrical power of different countries around the world. Using the breakdown of electricity generation sources, an average emissions factor is calculated for each country. This factor is then used to estimate the carbon footprint associated with electricity consumption in countries around the world.

Supplementary Fig. 6 provides a more detailed breakdown of electricity generation for each country [7, 8]. Using the energy mix for a given country, an average impact factor for electricity ($f_{electricity}$) is calculated and multiplied by the sum of the energy used to power the stand mixer ($E_{Fabrication}$) and the energy used to desiccate the composite ($E_{desiccation}$). For this LCA, an electrical grid efficiency (η) of 90% is assumed.

Solvent use during manufacture of Engineered Biopolymer Composite

The impact of solvent use, $Solvent_{Imp.}$ (Equation 4), is the solvent used to initially dissolve lignin during the manufacture of the lignin biocomposite. Equation 2 shows the general form for calculating the impact of the solvent, which considers the environmental impact of lignin-based BSC when either DMSO or a mixture of acetic acid and water is used as a solvent. However, the impact of solvent can be significantly reduced – or even avoided – through solvent recapture, represented by the percentage of recapture ($\%recapture$).

$$Solvent_{Imp.} = (m_{water} * f_{water} + m_{acid} * f_{acid}) * (1 - \%recapture) \quad (4)$$

where m_{water} and m_{acid} represent the masses of water and acid, respectively; f_{water} and f_{acid} are the corresponding emission factors for water and acid; and $\%recapture$ is the solvent recapture efficiency.

Environmental impact allocated to biopolymers used for EBC manufacture

The impact from biopolymer use, $Biopolymer_{Imp.}$ (Equation 5), accounts for both the environmental impacts associated with the production of the biopolymer constituents and the environmental benefit of biogenic carbon sequestration within the Engineered Biopolymer Composite. In this study, the biopolymer system consists primarily of kraft lignin and methyl cellulose (MC). The footprint associated with biopolymer production is calculated from the mass and emission factor of each component. For lignin, the production footprint is additionally multiplied by an allocation factor, L_{fac} , to account for varying allocation approaches related to lignin production. Although lignin is often treated as a waste by-product of pulping processes and may therefore be assigned little or no environmental burden, increased commercial demand for lignin-based materials could shift lignin toward co-product or primary-product status. Accordingly, allocation scenarios are included in this LCA to better capture the range of possible environmental outcomes.

For the primary analysis presented in this study, the full environmental burden associated with lignin production was allocated to lignin itself. Additional LCA scenarios, however, were also evaluated using reduced allocation factors in order to demonstrate the potential environmental benefits achievable under alternative allocation approaches. These allocation scenarios are represented through L_{fac} in Equation 5.

In addition to production impacts, the incorporation of carbon-rich biopolymers into the composite enables long-term biological carbon storage, representing a key environmental benefit of the material system [4]. Therefore, a carbon sequestration term, $Biopolymer_{sequestered CO_2}$, is subtracted from the total biopolymer impact to account for the amount of biogenic carbon retained within the composite structure.

$$Biopolymer_{Imp.} = f_{Lignin} * m_{Lignin} * L_{facf_{MC}} * m_{MC} - Biopolymer_{sequestered\ CO_2} \quad (5)$$

Computing the carbon footprint worldwide

This formulation is used to compute the global carbon footprint map presented in this study. Each term in Equation 6 directly corresponds to the aggregated components defined in Equation 1, enabling spatially resolved life cycle assessment results across different geographical regions.

Transportation impacts are calculated from lignin and cellulose transport distances, weighted by material mass fractions and normalized by energy efficiency assumptions. Electricity impacts combine baseline process electricity intensity with country-specific energy factors. Solvent impacts account for acetic acid usage and recovery efficiency. The binder term represents the biopolymer contribution from lignin-derived material inputs, while the final negative term represents biogenic carbon sequestration associated with carbon stored in the methyl cellulose and lignin matrix.

$$\begin{aligned} \text{Total Impact} = & \underbrace{\left(\frac{distances_L \cdot 0.148}{1000} \cdot 0.9 + \frac{distances_C \cdot 0.148}{1000} \cdot 0.1 \right) \cdot 0.196}_{Transportation_{Imp.}} \\ & + \underbrace{0.196 + 0.0597 \cdot Energy_Factors}_{Electricity_{Imp.}} + \underbrace{0.143 \cdot 1.184 \cdot 0.85}_{Solvent_{Imp.}} \\ & + \underbrace{0.10 \cdot 0.118 \cdot 0.711 \cdot 1.0}_{Binder\ (Biopolymer\ input)} + \underbrace{-2.2 \cdot 0.104 - 1.6 \cdot 0.0444}_{Biopolymer_{sequestered\ CO_2}} \end{aligned} \quad (6)$$

Life cycle inventory for analysis

To compute the average life cycle scenario for the manufacture of EBC, representative global transportation and processing assumptions were applied to the material inventory. The most frequent transportation distance observed for both lignin and methyl cellulose supply chains was assumed to be approximately 1000 km. In addition, a transport distance of 1000 km was assumed for the delivery of acetic acid to the manufacturing facility. These transportation assumptions were combined to determine the transportation demand expressed in tonne-kilometers (tkm) used in the LCA inventory.

For the baseline global scenario, lignin was conservatively assigned the full environmental burden associated with its production in order to represent a high-impact allocation case. This assumption provides an upper-bound estimate for the environmental footprint of the biopolymer system, while additional scenarios with reduced lignin allocation factors were also evaluated separately. Furthermore, an overall manufacturing energy efficiency of 90% was assumed to account for process-level

Supplementary Table S2: Life Cycle Assessment (LCA) inputs used for the average global scenario in Figures.

Parameter	Value	Units
Density of EBC	1805	kg/m ³
Density of concrete	2400	kg/m ³
EBC compressive strength	35	MPa
Concrete compressive strength	20	MPa
Energy efficiency factor	0.9	–
Scenario Definitions		
Scenario 1: 15% re-capture, lignin waste product		
solvent re-capture	0.15	–
Percent of biopolymer binder that is methyl cellulose	0.1	–
Scenario 2: 0% recapture, lignin main product		
Percent recapture	1.0	–
Lignin allocation factor	1.0	–
Material and Energy Inputs for 1 Kg of EBC		
Acetic acid	0.10714275	kg
Electricity (oven)	0.004622222	kWh
Electricity (mixing)	0.048148148	kWh
Total electricity	0.05277037	kWh
Lignin	0.148148	kg
Transportation	0.255340298	tkm
Water	0.0357143	kg

energy optimization during EBC production. The resulting material and energy inventory inputs used for the average global manufacturing scenario are summarized in Table S2.

Supplementary Table S3: Comparison of environmental impact categories for EBC and conventional concrete on a volumetric basis and strength-normalized basis.

Impact Category	per m ³		per m ³ MPa	
	EBC	Concrete	EBC	Concrete
Carcinogens	30.2666	48.7224	0.864761	2.43612
Non-carcinogens	17.0764	75.5400	0.487898	3.77700
Respiratory inorganics	0.637936	1.0488	0.018227	0.05244
Ozone layer depletion	3.52×10^{-5}	0.000222	1.01×10^{-6}	1.11×10^{-5}
Respiratory organics	0.454400	0.8784	0.012983	0.04392
Aquatic ecotoxicity	102936.0324	562466.2	2941.029	28123.31
Terrestrial ecotoxicity	8720.4973	195279.5	249.1571	9763.974
Terrestrial acidification/nutritification	12.7250	32.1816	0.363571	1.60908
Land occupation	41.1521	591.2808	1.175774	29.56404
Aquatic acidification	3.69814	5.5800	0.105661	0.27900
Aquatic eutrophication	0.109274	0.2856	0.003122	0.01428
Global warming	-222.5114	501.6977	-6.35747	25.08489

References

- [1] Miao, B. H. *et al.* Development of biopolymer composites using lignin: A sustainable technology for fostering a green transition in the construction sector. *Cleaner Materials* 100279 (2024).
- [2] Miao, B. H., Woo, D., Lesh, A. C., Loftus, D. J. & Lepech, M. D. Solvent selection enables sustainable and affordable lignin biocomposite for cement-free construction. *Cleaner Materials* 100371 (2026).
- [3] Ma, Q., Wang, L., Zhai, H. & Ren, H. Lignin dissolution model in formic acid–acetic acid–water systems based on lignin chemical structure. *International Journal of Biological Macromolecules* **182**, 51–58 (2021).
- [4] Miao, B. H. *et al.* Life cycle assessment and design of lignoblock: A lignin bound block on the path towards a green transition of the construction industry. *Journal of Cleaner Production* 143610 (2024).
- [5] Joliet, O. *et al.* Impact 2002+: a new life cycle impact assessment methodology. *The international journal of life cycle assessment* **8**, 324–330 (2003).
- [6] Peng, Y. *et al.* Mycorrhizal association and life form dominantly control plant litter lignocellulose concentration at the global scale. *Frontiers in Plant Science* **13**, 926941 (2022).
- [7] Ritchie, H. & Rosado, P. Electricity mix. *Our World in Data* (2020).
<https://ourworldindata.org/electricity-mix>.
- [8] (IEA), I. E. A. Energy system of africa. <https://www.iea.org/regions/africa> (2023). Accessed: 2024-12-06.